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暨全国代表大会

Sound waves from primordial black hole formations

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Outline

01

Introduction

02

Simulation Setup

03

Numerical Results

04

Conclusions



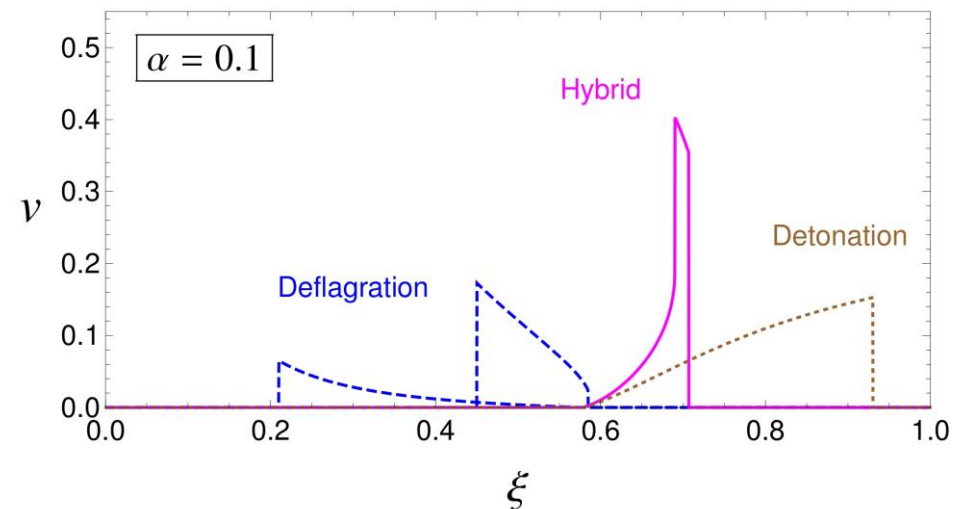
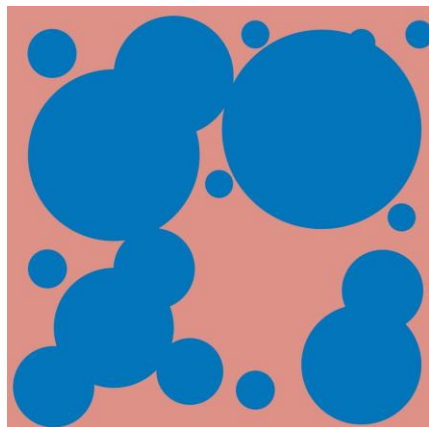
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01

Introduction

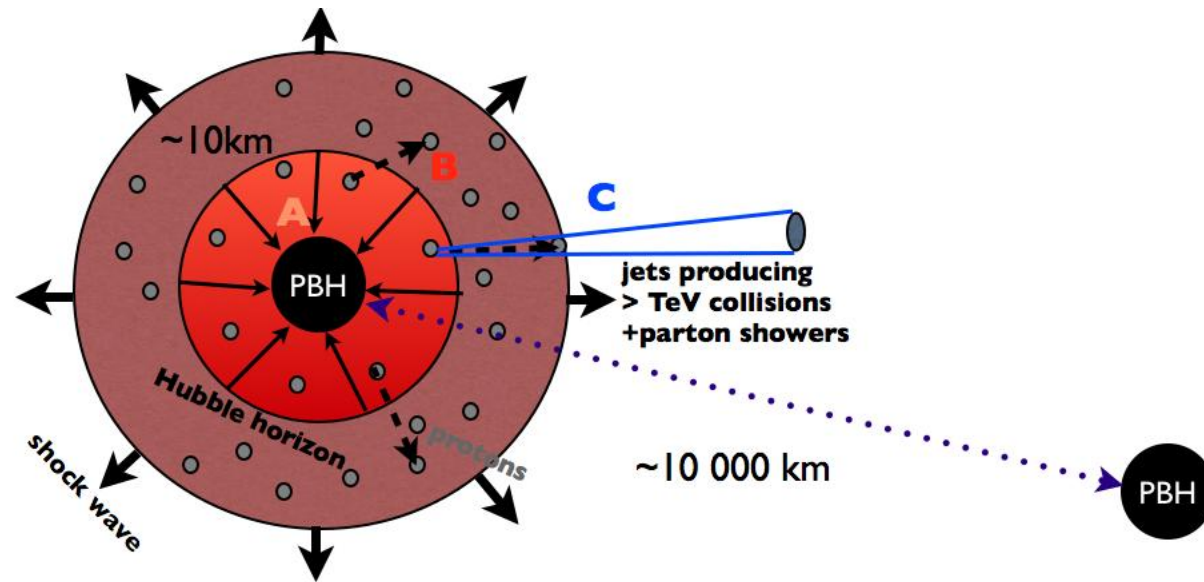
01 | Introduction

- Sound waves can be generated and propagate in the early Universe
- Cosmological first-order phase transitions:
 - **Wall-fluid interaction** generates sound waves
 - Semi-analytically derived **self-similar velocity profiles**
 - Serve as **initial conditions** for the **sound shell model** to semi-analytically calculate their gravitational wave spectra [1608.04735,1909.10040,2007.08537]



01 | Introduction

- Primordial black hole (PBH) formation also generate sound waves!
- Collisions between these sound waves are expected to generate gravitational waves as well
- Profiles of PBH-induced sound waves are governed by nonlinear dynamics of gravitational collapse
- **Need nonlinear numerical simulation!**



[1904.02129,1904.11482]



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02

Simulation Setup

02 | Misner-Sharp formalism

- PBH formation from curvature perturbations

- Metric ansatz: $ds^2 = -A(t, r)^2 dt^2 + B(t, r)^2 dr^2 + R(t, r)^2 d\Omega^2$

- Energy-momentum tensor: $T_{\mu\nu} = (\rho + p)u_\mu u_\nu + pg_{\mu\nu}$

- Equation of state: $p = \omega\rho$

$$D_r A = \frac{-A}{\rho + p} D_r p,$$

$$D_r M = 4\pi\Gamma\rho R^2,$$

$$\Gamma = \sqrt{1 + U^2 - \frac{2M}{R}},$$

$$D_t R = U,$$

$$D_t U = - \left[\frac{\Gamma}{(\rho + p)} D_r p + \frac{M}{R^2} + 4\pi R p \right],$$

$$D_t M = -4\pi R^2 U p,$$

$$D_t \rho = -\frac{(\rho + p)}{\Gamma R^2} D_r (U R^2).$$

- Break down due to the singularity formation

- Need to excise the black hole interior region

02 | Cosmological setup

- To enhance numerical accuracy and stability, we **factor out the background evolution** from the dynamical variables

$$\xi \equiv \ln(t/t_0)$$

$$\bar{r} \equiv r/R_H$$

$$R_H = H^{-1}(t_0) = t_0/\alpha$$

$$\rho = \rho_b \bar{\rho},$$

$$p = \rho_b \bar{p},$$

$$R = R_b \bar{R} = a r \bar{R},$$

$$U = H R \bar{U},$$

$$M = \frac{4\pi}{3} \rho_b R^3 \bar{M},$$



$$A = \bar{\rho}^{-3\alpha\omega/2}, \quad (11a)$$

$$\bar{\rho} = \bar{M} + \frac{\bar{r} \bar{R}}{3(\bar{r} \bar{R})'} \bar{M}', \quad (11b)$$

$$\bar{\Gamma}^2 \equiv \frac{\Gamma^2}{a^2 H^2 R_H^2} = e^{2(1-\alpha)\xi} + \bar{r}^2 \bar{R}^2 (\bar{U}^2 - \bar{M}), \quad (11c)$$

$$\partial_\xi \bar{R} = \alpha \bar{R} (\bar{U} A - 1), \quad (11d)$$

$$\begin{aligned} \partial_\xi \bar{U} = \bar{U} - \alpha A \left[\bar{\Gamma}^2 \frac{\bar{p}'}{\bar{r} \bar{R} (\bar{R} + \bar{r} \bar{R}') (\bar{\rho} + \bar{p})} \right. \\ \left. + \frac{1}{2} (2\bar{U}^2 + \bar{M} + 3\bar{p}) \right], \end{aligned} \quad (11e)$$

$$\partial_\xi \bar{M} = 2\bar{M} - 3\alpha \bar{U} A (\bar{p} + \bar{M}), \quad (11f)$$

$$\begin{aligned} \partial_\xi \bar{\rho} = -\alpha (\bar{\rho} + \bar{p}) A \left(3\bar{U} + \frac{\bar{r} \bar{R} \bar{U}'}{\bar{R} + \bar{r} \bar{R}'} \right) + 3\alpha (1 + \omega) \bar{\rho}, \end{aligned} \quad (11g)$$

02 | Numerical implementation

- Super-horizon initial conditions: long-wavelength approximation
- Spatial perturbations are encoded in $K(r)$:

$$ds^2 = -dt^2 + a^2(t) \left(\frac{dr^2}{1 - K(r)r^2} + r^2 d\Omega^2 \right)$$

- A Gaussian profile in this work:

$$K(r) = \mathcal{A}\bar{K}(r) = \mathcal{A}e^{-(r/r_m)^2}$$

- Critical threshold: $\delta_c \approx 0.49774$
- Spatial direction: 4th-order central difference + logarithmic grid
- Time direction: 4th-order Runge-Kutta method



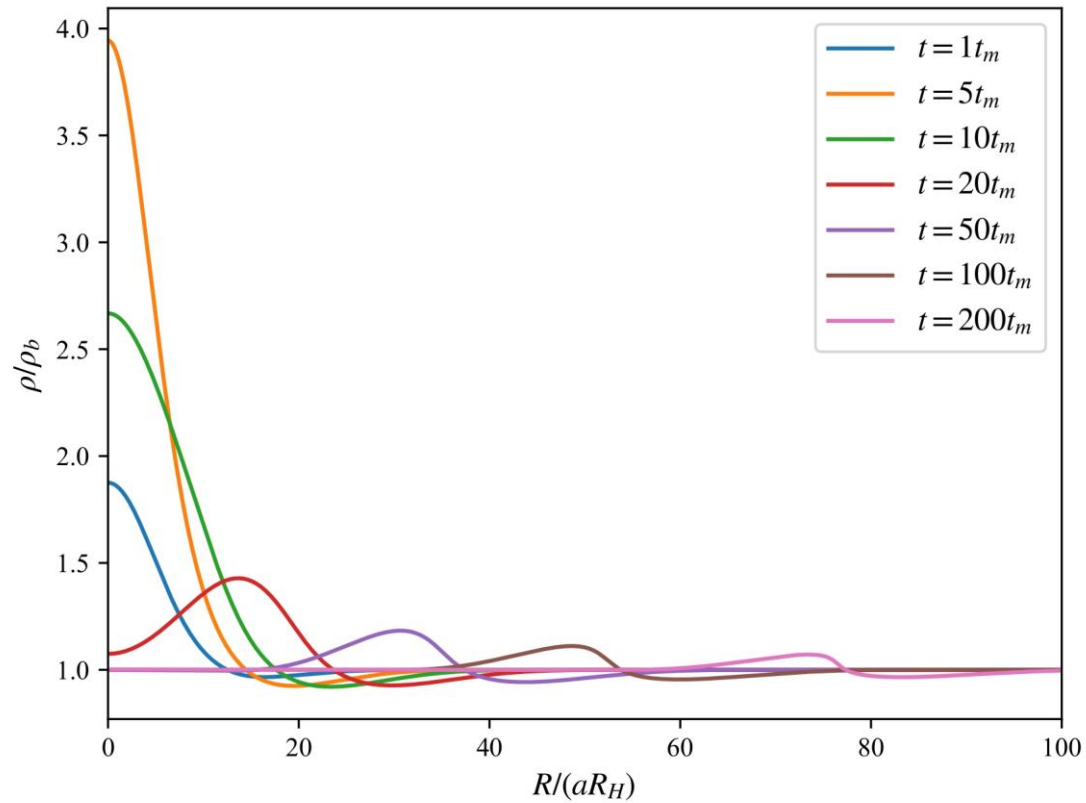
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03

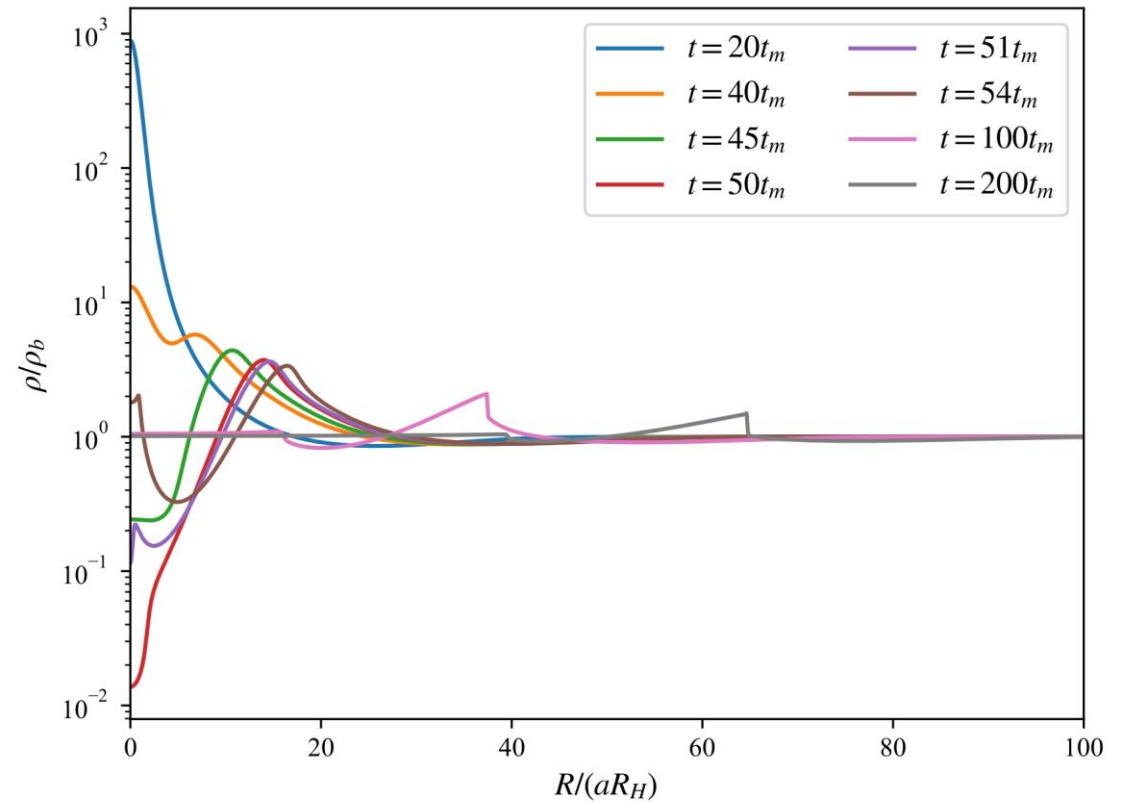
Numerical Results

03 | Subcritical cases

- No black hole is formed ($\omega = 1/3$)



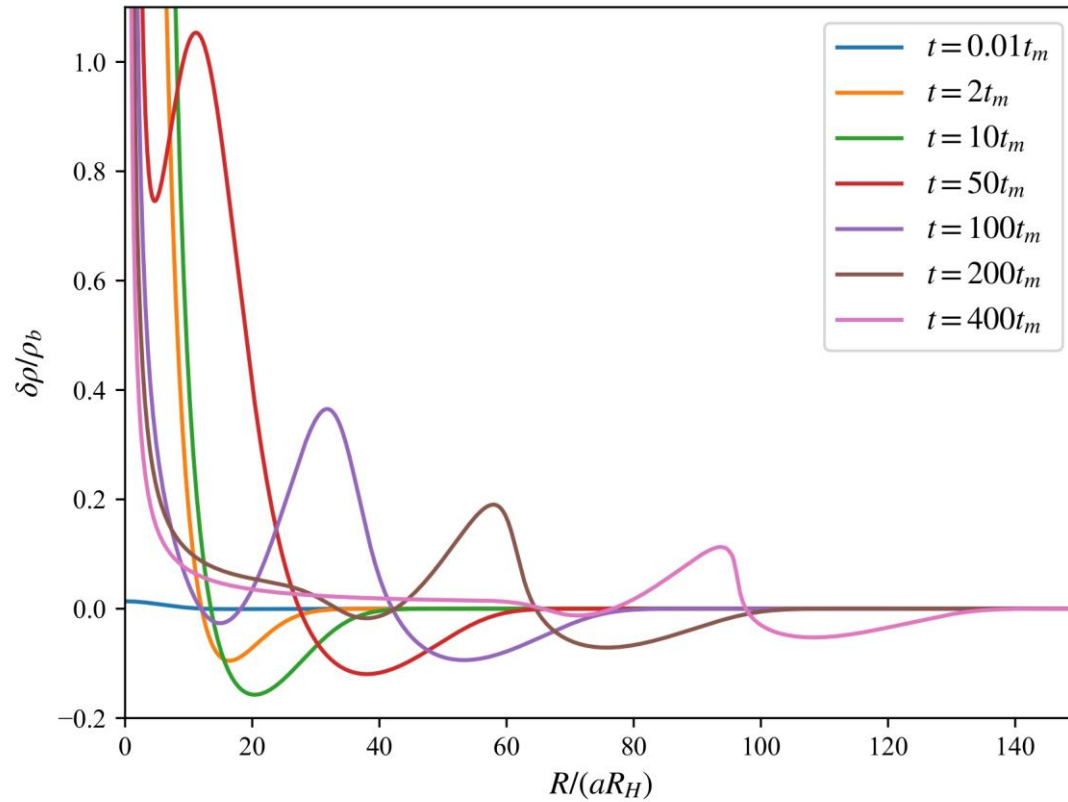
$$\delta_m = 0.3 \ll \delta_c$$



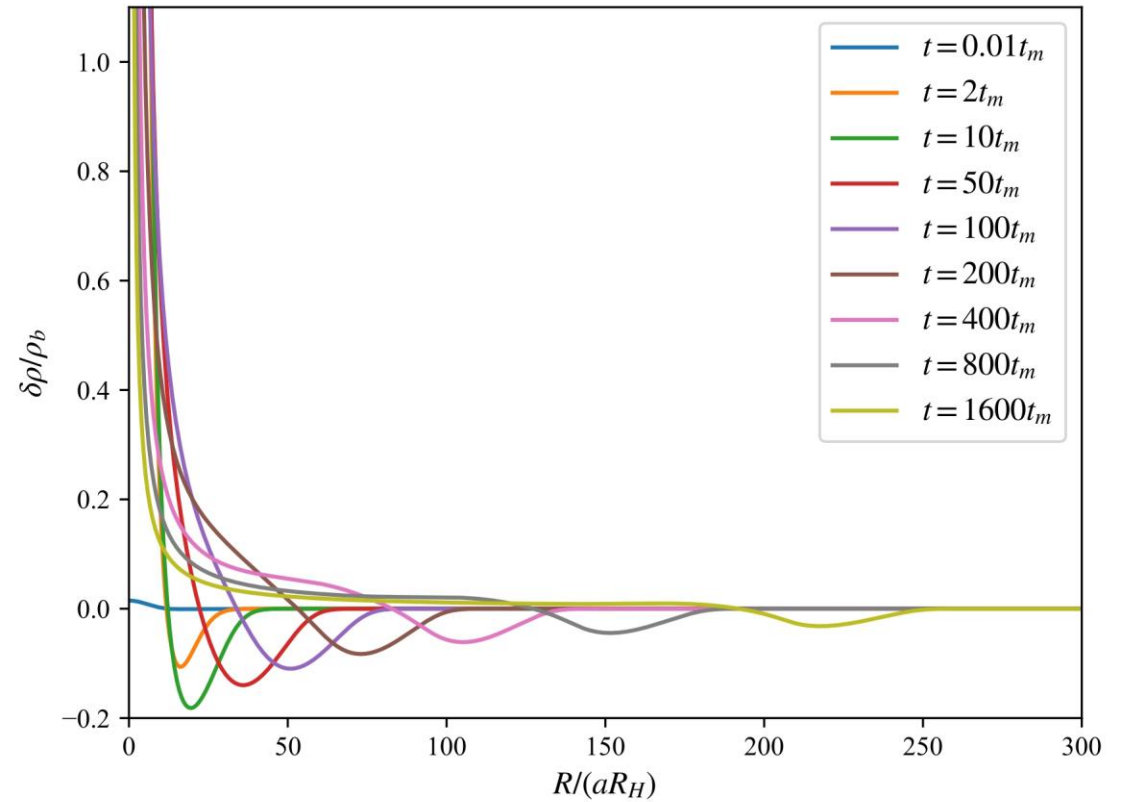
$$\delta_m = 0.495 \lesssim \delta_c$$

03 | Supercritical cases

- The PBH is formed ($\omega = 1/3$)



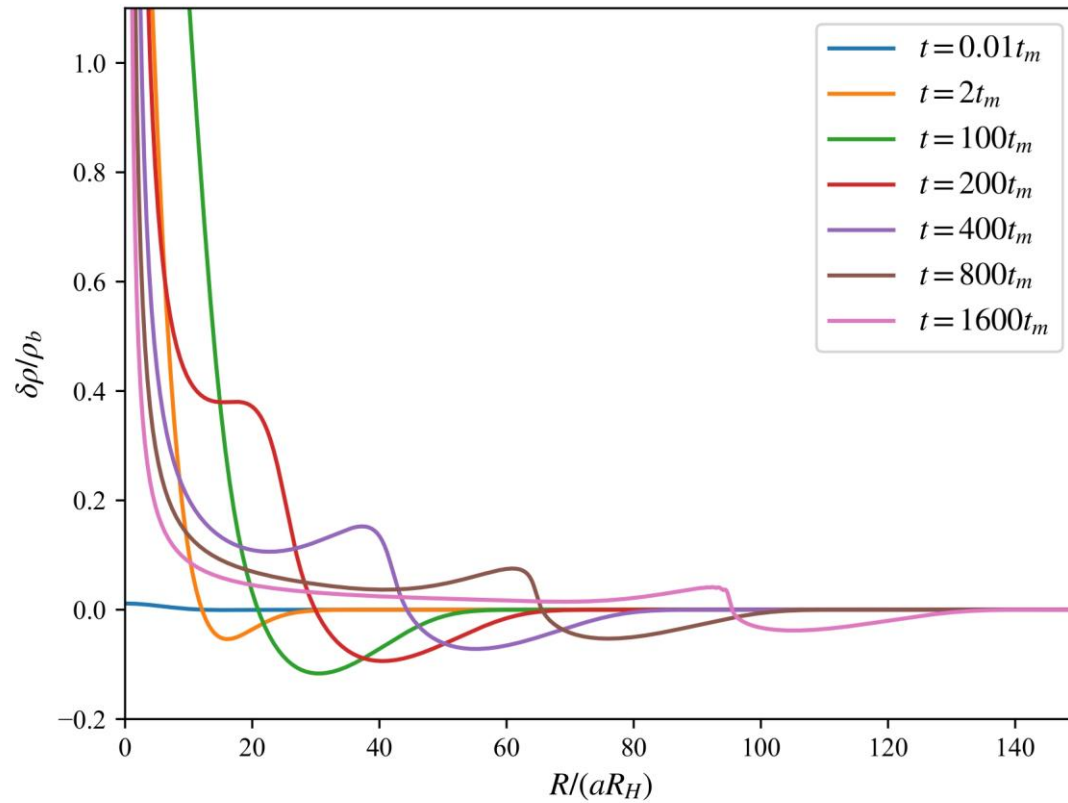
$$\delta_m = 0.5 \gtrsim \delta_c$$



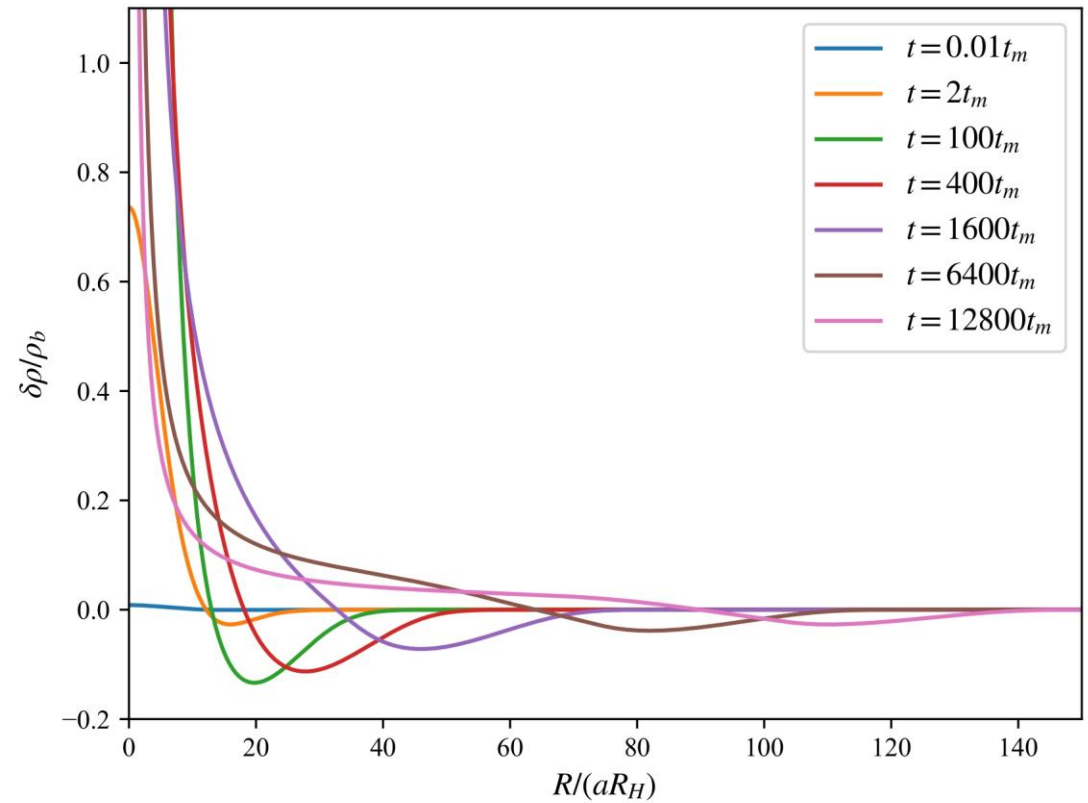
$$\delta_m = 0.55 > \delta_c$$

03 | Softer equations of state

- $\delta_m - \delta_c \approx 0.00226$



$$\omega = 1/5$$



$$\omega = 1/10$$

03 | Propagation of sound waves

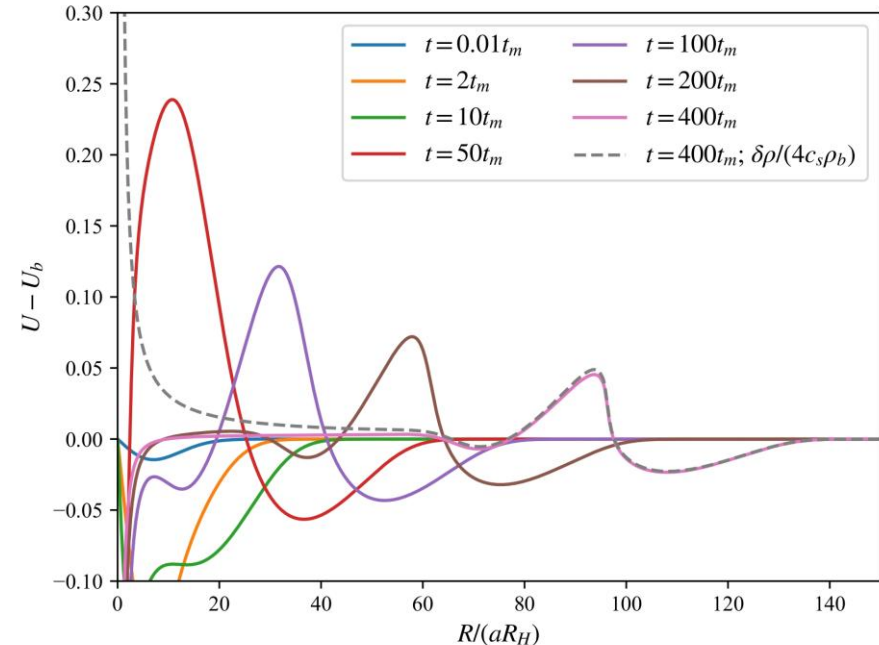
- Once these waves move sufficiently far from the PBH, they can be treated as **linear perturbations** on a flat FLRW background, i.e., sound waves

$$\begin{aligned} T^{00} &= \rho_b(1 + \delta) + (1 + \omega)\rho_b(1 + \delta)a^2 u^i u_i, \\ T^{0i} &= (1 + \omega)\rho_b(1 + \delta)u^i, \\ T^{ij} &= \omega \frac{h^{ij}}{a^2} \rho_b(1 + \delta) + (1 + \omega)\rho_b(1 + \delta)u^i u^j, \end{aligned}$$

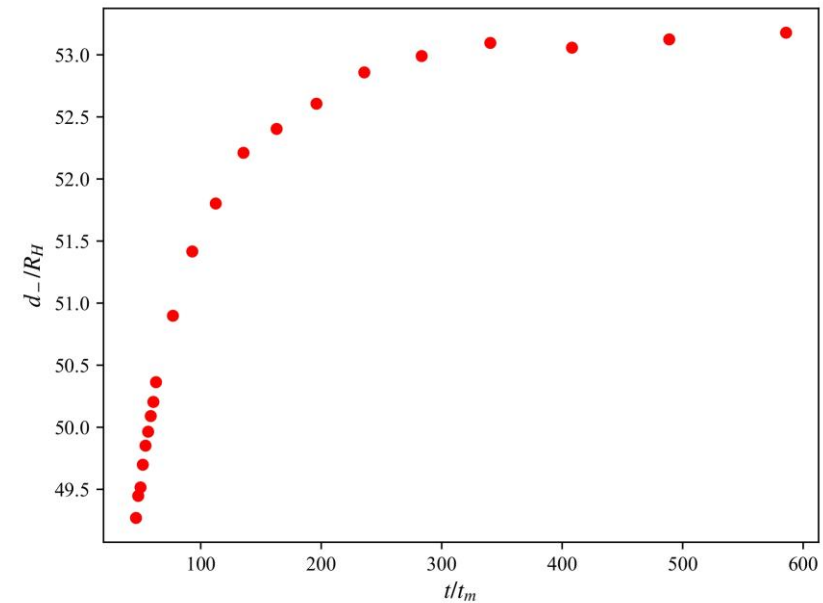
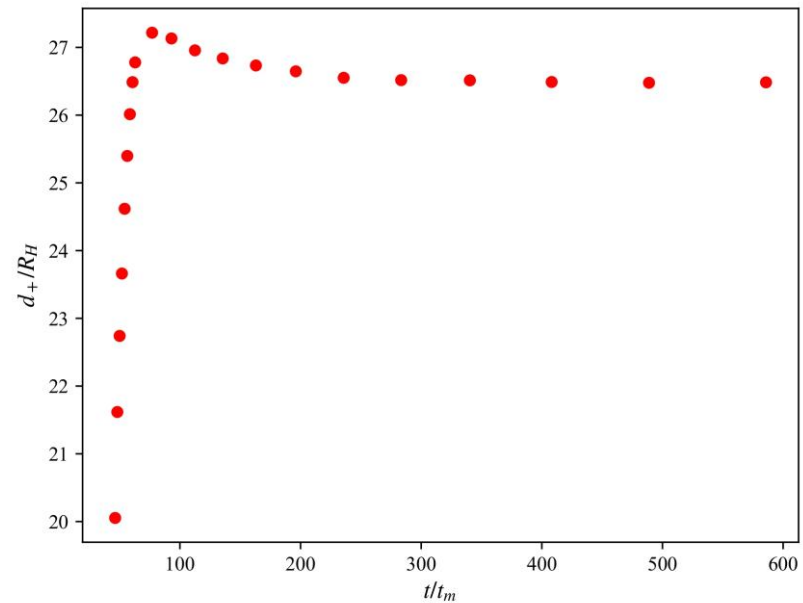
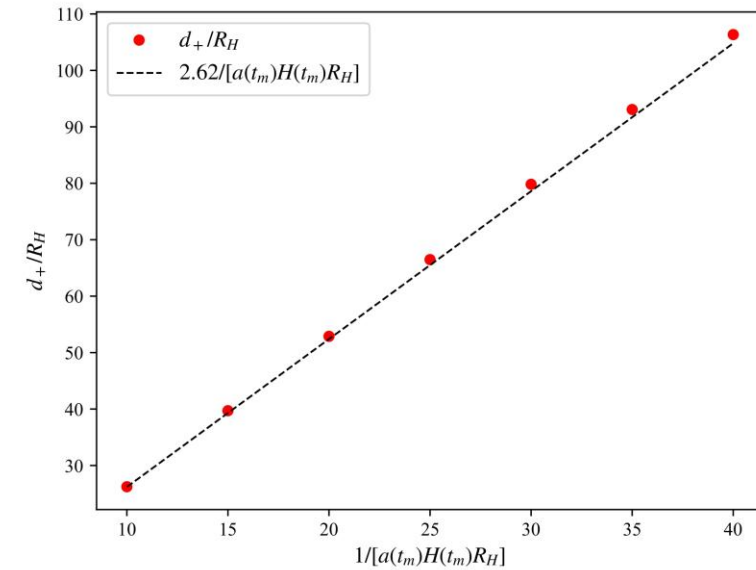
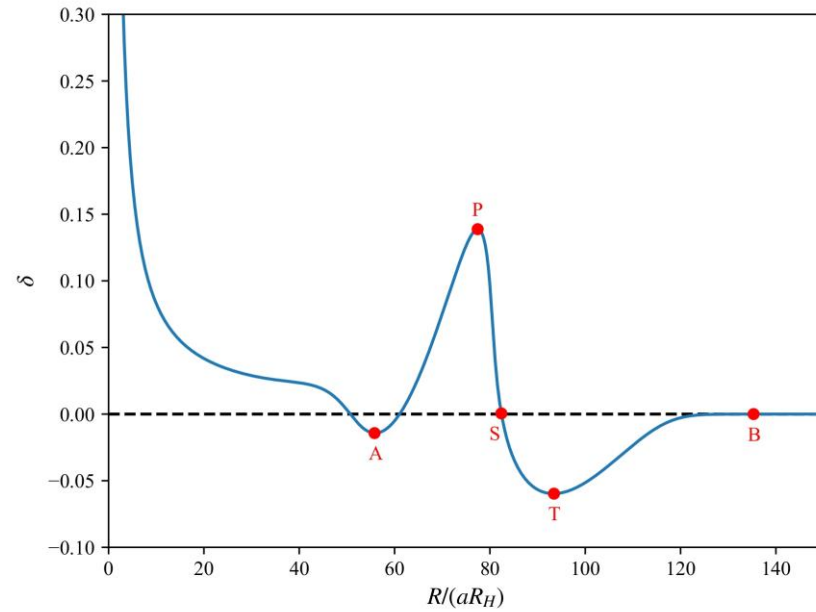


$$\begin{aligned} \dot{\delta} + (1 + \omega) \left(u' + \frac{2u}{r} \right) &= 0, \\ (1 + \omega)\dot{u} + \omega \frac{\delta'}{a^2} + (2 - 3\omega^2 - \omega) \frac{\dot{a}}{a} u &= 0, \end{aligned}$$

$$\begin{aligned} \delta &= -4a\partial_t(a\phi) = 4c_s \frac{f'(r - c_s R_H a)}{r}, \\ u &= \phi' = \frac{f'(r - c_s R_H a)}{ar} - \frac{f(r - c_s R_H a)}{ar^2} \\ &\approx \frac{f'(r - c_s R_H a)}{ar} = \frac{\delta}{4ac_s}, \\ U - U_b &\approx \frac{\delta\rho/\rho_b}{4c_s} \end{aligned}$$

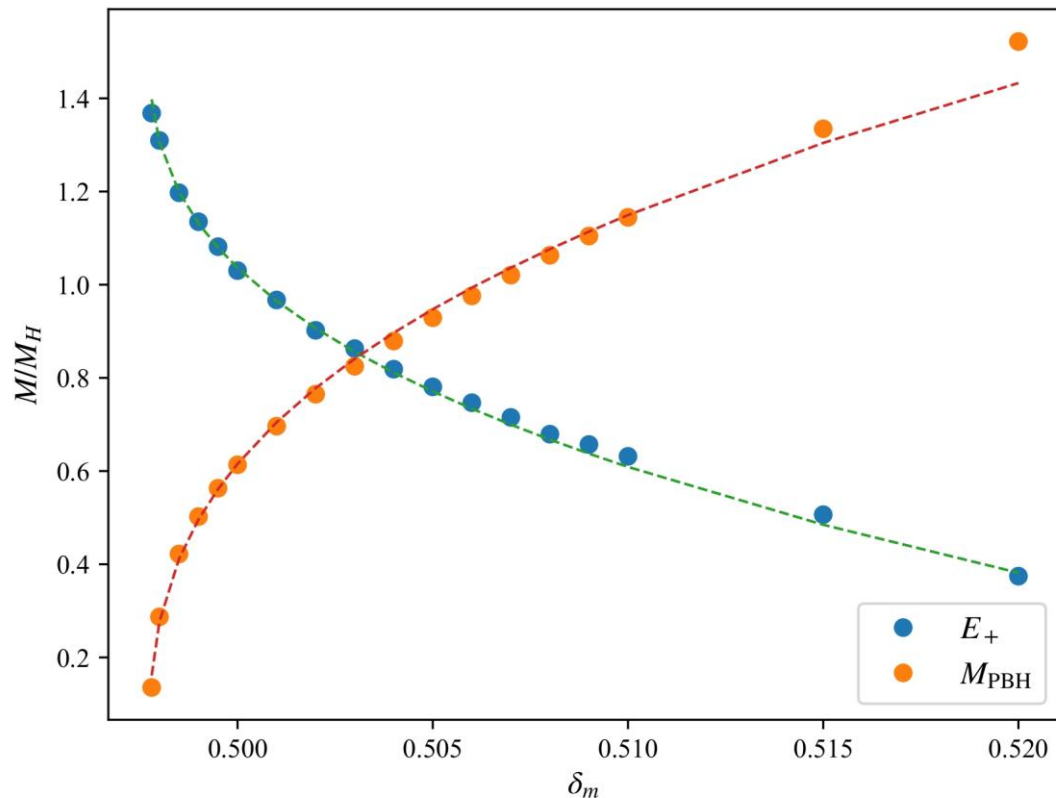


03 | Thickness of sound shells



03 | Energy of sound shells

- PBH mass follows the scaling law
- We find that the energy of overdense shells follows a **similar scaling law**



$$M_{\text{PBH}} = M_H \mathcal{K} (\delta_m - \delta_c)^\sigma$$

$$E_+ = M_H [\mathcal{K}_+ (\delta_m - \delta_c)^{\sigma_+} + C_+]$$

the fit yields $\sigma \approx \sigma_+ \approx 0.37$, $\mathcal{K} \approx 5.84$,

$\mathcal{K}_+ \approx -4.70$, and $C_+ \approx 1.52$.



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04

Conclusions

04 | Conclusions and discussion

- Near-critical perturbations produce a compression wave with **both overdense and underdense shells**
- Significantly supercritical perturbations yield **only an underdense shell**
- A softer equation of state suppresses the formation of compression waves
- The comoving thickness of sound shells **remains nearly constant** during propagation and **scales with the comoving Hubble radius at horizon re-entry**
- Energy of the overdense shells follows a scaling law analogous to that of the PBH mass
- Outlook
 - more complex curvature perturbations, such as multi-peak or oscillatory profiles
 - non-spherical effects
 - Other PBH formation mechanisms

Thank You!